Autumn 2024 – DPS – END SEM

Level : 1	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH1101	Mechanics I	Theory	Golam Mortuza Hossain
Level : 2	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH2101	Waves and optics	Theory	Rangeet Bhattacharyya
2	PH2102	Mechanics II	Theory	Ananda Dasgupta
Level : 3	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH3101	Classical Mechanics	Theory	Rajesh Kumble Nayak
2	PH3102	Quantum Mechanics	Theory	Siddhartha Lal
3	PH3103	Mathematical Methods of Physics	Theory	Koushik Dutta
4	PH3104	Electrical Circuits and Electronics	Theory	Dhananjay Nandi
Level : 4	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH4101	Condensed Matter Physics	Theory	Subhasis Sinha
2	PH4102	Introductory Astrophysics	Theory	Sudip Kumar Garain
3	PH4104	Nonlinear Dynamics	Theory	Anandamohan Ghosh
4	PH4105	Advanced Mathematical Methods of Physics	Theory	Ananda Dasgupta
5	PH4106	Field Theory and Relativistic Quantum Mechanics	Theory	Ritesh Kumar Singh
6	PH4107	Advanced Electricity, Magnetism, and Optics	Theory	Nirmalya Ghosh
7	PH4110	Soft Condensed Matter Physics	Theory	Arindam Kundagrami
Level : 5	Programme : MS, MP, MR, IP, RS			
Sl. No.	Course Co	Course Name	Type	Instructor(s)
1	PH5103	Advanced Condensed Matter Physics	Theory	Amit Ghosal

Indian Institute of Science Education and Research Kolkata
Department of Physical Sciences

PH1101 :: Mechanics I
End-semester Examination (Autumn 2024)
Marks: 50, Duration: 2 hours 30 minutes

1. Compute the following limits: [2+2]

$$(i) \quad \lim_{y\to 1} \frac{y^3-1}{y-1} \ , \quad (ii) \quad \lim_{x\to 0+} x\log x \ .$$

2. Find the locations and the corresponding values of the finite maxima and minima of the potential function $U(x) = 2x^3 - 6x$. Sketch the graph of the potential and mark these extrema in the graph. [5+2]

3. Are these two vectors $u = ax$ and $v = ax + b$ linearly independent? Answer with proper justifications, if a and b are arbitrary constants. [3]

4. The orbital eccentricity of Phoebe, an irregular moon of the planet Saturn, is 0.16. In its elliptical orbit, Phoebe reaches as close as 10.9×10^6 km to Saturn's center. Find the semi-major and semi-minor axes of the orbit. What is maximum possible distance of Phoebe from the center of Saturn? [4+2]

5. Show that the total linear momentum of an isolated system of N interacting particles is a conserved quantity. [6]

6. Consider a system of two particles having mass $3m$ and m. Find the position of the center of mass if positions of the particles are given by $\vec{r}_1 = \cos(\omega t)\hat{i} + \sin(\omega t)\hat{j}$ and $\vec{r}_2 = \cos(\omega t)\hat{i} - \sin(\omega t)\hat{j}$ respectively. [4]

7. Find the general solution for the position $x(t)$ of a particle whose motion is described by the equation $\ddot{x}(t) = 2\dot{x}(t) - 17x(t)$. If the particle satisfies the initial conditions $x(0) = 0$ and $\dot{x}(0) = 2$ then find the position of the particle at a time t and also sketch the position $x(t)$. [5+3+2]

8. Show that the orbit of planet Mercury of mass M moving around the Sun having mass $M_\odot$ is an ellipse. In polar coordinates, the acceleration vector is given to be $\vec{a} = (\ddot{r} - r\dot{\theta}^2)\ \hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\ \hat{\theta}$. [10]

G. M. Hossain

PART A: Answer any **one** question

A-1. A wave travels along a string with a speed of **150** m/s when the tension is **75** N. Calculate the linear mass density (mass per unit length) of the string. [2]

A-2. Show that multiplication of a complex number z by $e^{i\theta}$ represents a positive rotation of the vector by angle θ, without changing its magnitude. [2]

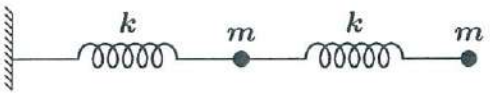
PART B: Answer any **six** questions

B-1. Consider the following spring-mass system consisting of two springs and two masses. [8]

(a) Derive the equations of motion. [3]

(b) Solve for the normal mode frequencies of the system. [3]

(c) What is the ratio of amplitudes of the two masses when the system oscillates with lower normal mode frequency? [2]



B-2. Consider a damped harmonic oscillator with unit mass where the damping coefficient $\gamma(\omega)$ depends on the driving frequency ω as $\gamma(\omega) = \alpha\omega$, where α is a constant. [8]

(a) Write the equation of motion. [2]

(b) Derive an expression for the steady-state amplitude $A(\omega)$ and discuss how it differs from constant damping. [3+1]

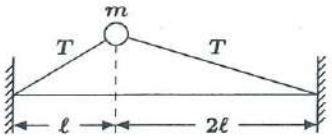
(c) What is the physical significance of the resonance peak shifting due to $\gamma(\omega)$? [2]

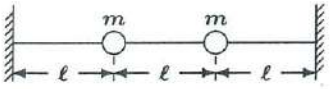
B-3. A string of length 3ℓ and negligible mass is attached to two fixed supports at its ends. The tension in the string is T. [8]

(a) A particle of mass m is attached at a distance ℓ from one end of the spring, as shown. Set up the equation for a small transverse oscillation of m, and find the period. [3]

(b) An additional particle of mass m is connected to the string as shown, dividing it into three equal segments each with tension T. Sketch the appearance of the string and masses[1] in the two separate normal modes of transverse oscillations. [4]

(c) Calculate ω for the normal mode which has the higher frequency.





B-4. Consider an electromagnetic wave traveling through a medium of refractive index n_1. [8]

(a) Show that Maxwell's equations imply that the wave must be transverse in nature. [2]

(b) Derive and show how the electric and magnetic field vectors are connected. [2]

Suppose the wave is obliquely incident on a plane boundary separating two media having refractive indices n_1 and n_2.

(c) Derive the law of reflection. [2]

(d) Derive the law of refraction. [2]

B-5. Suppose the wave is obliquely incident on a plane boundary separating two media having refractive indices n_1 and n_2. The light is going from n_1 medium to n_2 medium. [8]

(a) Write Fresnel's equations for p-type and s-type polarizations. [2]

Ranjan Bhattacharyya
12/12/24

(b) ~~Using the above, show that Brewster's angle can be written as $\tan\theta_B = n_2/n_1$ for p-polarization.~~ show that at Brewster's angle $\theta_I + \theta_T = \pi/2$. [3]

(c) Show that Brewster's angle does not exist for s-type polarization. [3]

B-6. Consider two thin slits of width D, kept d distance apart and a screen placed at L distance from the point sources. The sources emit monochromatic light of wavenumber $k = \frac{2\pi}{\lambda}$. The intensity on a screen at an angle θ is given by [8]

$$I(\theta) = I_o \cos^2\left(\frac{kd\sin\theta}{2}\right)\mathrm{sinc}^2\left(\frac{kD\sin\theta}{2}\right)$$

where, $\mathrm{sinc}(\phi) = \frac{\sin\phi}{\phi}$.

It is given that $L = 1$ m, $d = 1.2$ mm, and $\lambda = 500$ nm.

(a) The 5th maximum of the double slit pattern is missing on either side of the central maximum. Find the value of D. [4]

(b) Calculate the distance between two successive maxima. [2]

(c) Plot the intensity as a function of x for the above value of D. [2]

B-7. Suppose you have a beam of light (of unknown type) and a good supply of optical elements, such as, glass plates, polarizers, half-wave and quarter-wave plates, photo-detectors. Design suitable experimental protocols to find whether the incident beam is [8]

(a) linearly polarized, [2]

(b) right circularly polarized, [2]

(c) left circularly polarized, [2]

(d) unpolarized. [2]

B-8. Consider a denser to rarer planer interface. The refractive indices of the two media are given by n_1 and n_2. [8]

(a) Starting from Fresnel's equations, show that Brewster's angle θ_B could be given by $\tan\theta_B = \frac{n_2}{n_1}$. [4]

(b) Calculate the critical angle for total internal reflection for $n_1 = 3/2$ and $n_2 = 1$. [2]

(c) Calculate the Brewster's angle for $n_1 = 3/2$ and $n_2 = 1$. [2]

Ranjan Bhattacharya
12/12/24

PH2102 End Semester Examination

Full marks : 50 Time : **150 minutes**

Group - A

Answer any two

Q 1) a) Show that the velocity 4-vector U^μ has a constant norm $U^\mu U_\mu$.

b) Calculate the value of $U^\mu A_\mu$ - where A^μ is the acceleration 4-vector.

c) Using the velocity 4-vector show that under a special Lorentz transformation we have

$$\frac{\gamma(u')}{\gamma(u)\,\gamma(v)} = 1 - \frac{u_x v}{c^2}$$

where the symbols have their usual meanings.

d) Alice, standing beside a railway track, throws a ball at a speed u towards a wall a distance D away from her. Meanwhile, Bob is riding a train that is moving in the direction parallel to the path of the ball with a speed v with respect to Alice. Consider the two events - Alice throwing the ball and the ball hitting the wall.

(i) Determine the proper time between the two events in terms of D and u.

(ii) If the speed of the ball with respect to Bob is u', determine the time interval between these events, as measured by Bob, in terms of D, u' and v.

(iii) Use the relation $u' = \dfrac{u - v}{1 - \frac{uv}{c^2}}$, and the results in parts (d-i) and (d-ii) to give an alternative derivation of the result in part (c) above.

$(2 + 3 + 3 + (2 + 2 + 3))$

Q 2) a) A particle of rest mass M is moving at a speed u_0 decays into two bodies with equal rest mass m. Determine the speed of one of the decay particles with respect to the other particle.

b) A mother particle decays into three particles. The masses of the three decay particles are $m_0, 2m_0$ and $3m_0$. Their velocities after the decay are $\frac{3}{5}c\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, $\frac{4}{5}c\frac{\hat{i}-\hat{j}}{\sqrt{2}}$ and $\frac{3}{5}c\hat{k}$, respectively. What is the rest mass of the mother particle?

c) Show that the angle between the force acting on a point particle and its acceleration is always acute (i.e. $< 90°$) (5×3)

Q 3) A group $(G, *)$ is a mathematical structure comprising of a set G and a binary operation $* : G \times G \rightarrow G$ which obeys the following properties:

(i) Closure : $\forall g_1, g_2 \in G$, $g_1 * g_2 \in G$

(ii) Associativity : *i.e* $\forall g_1, g_2, g_3 \in G$: $(g_1 * g_2) * g_3 = g_1 * (g_2 * g_3)$

(iii) Identity : $\exists e \in G$: $e * g = g * e = g, \forall g \in G$

(iv) Inverse : $\forall g \in G$: $\exists g^{-1} \in G$: $g * g^{-1} = g^{-1} * g = e$

A Lorentz transformation can be written in the compact form $\mathbf{x}' = L\mathbf{x}$ (where $\mathbf{x}$ is a 4×1 column vector with elements $x^0 = ct, x^1 = x, x^2 = y$ and $x^3 = z$). Let $\Lambda^\mu{}_\nu$ represent the μ-th row, ν-th column element of the 4×4 matrix L. The matrix L obeys $L^T \eta L = \eta$.

a) Prove that the set of all Lorentz matrices $\mathcal{L} \equiv \left\{L : L^T \eta L = \eta\right\}$ form a group where the operation $*$ is matrix multiplication. *Matrix multiplication is always associative - so you only have to prove the other properties.*

b) Show that the subset $\mathcal{L}_+$ of all matrices L in $\mathcal{L}$ which obey $\det L = 1$ is a group under matrix multiplication as well.

c) Show that the subset $\mathcal{L}^\uparrow$ of all matrices L in $\mathcal{L}$ which obey $\Lambda^0{}_0 \geq 1$ is also a group under matrix multiplication

d) Prove that both $\mathcal{L}_+$ and $\mathcal{L}^\uparrow$ are proper subsets of $\mathcal{L}$, *i.e.* that there are elements in $\mathcal{L}$ which do not belong to either of the subsets. $(5 + 3 + 5 + 2)$

Group - B

Answer any one

Q 4) A light spring of unstretched length L and spring constant k has a small bob of mass attached to one end. The spring is suspended from the other end and the spring-bob system is allowed to swing in a vertical plane. The system is described by polar coordinates (r, θ), where r is the length of the spring and θ is the angle that the spring axis makes with the vertical.

a) Write down the Lagrangian for the system.

b) Use this to write down the Euler-Lagrange equations of motion.

c) At equilibrium both r and θ would be constant at r_0 and θ_0, respectively. From the equation of motion that you have derived above, determine the values of r_0 and θ_0.

d) For small oscillations, the departures $\rho \equiv r - r_0$ and $\phi \equiv \theta - \theta_0$ will be small. Rewrite the Euler-Lagrange equations of motion in terms of ρ and ϕ, only retaining terms of linear order in these variables and their derivatives.

e) Describe the motion of this system for small departures from equilibrium.

f) What is a conserved quantity for this system and why?

$(4 + 4 + 2 + 4 + 4 + 2)$

Q 5) a) Define cyclic coordinates. Give an example.

b) Prove that if a particular generalized coordinate is cyclic, then the corresponding generalized momentum is conserved.

c) Prove that the Jacobi function $h \equiv \sum_{i=1}^{n} p_i \dot{q}_i - L$ is conserved if the Lagrangian is explicitly independent of time.

d) The Lagrangian for a system with 1 degree of freedom is given by $L = \frac{1}{2}m\dot{q}^2 + Aq^2\dot{q}$.

i) Determine the Jacobi function for this system.

ii) If the particle is at q_1 at time t_1 and at q_2 at time t_2, determine the form of $q(t)$ between t_1 and t_2 by solving the Euler-Lagrange equation of motion for this system.

iii) Determine the value of the action for the actual path followed by the system from (q_1, t_1) to (q_2, t_2).

e) Show explicitly that the Euler-Lagrange equations of motion for the two Lagrangians L and $L' = L + \frac{df}{dt}$ are identical if f is an arbitrary smooth function of the generalized coordinates and time (but not of the generalized velocities). Can you explain why this is to be expected from the principle of least action? $(2+2+4+(2+2+3)+3+2)$

Important results and definitions

- $\gamma(u) = \frac{1}{\sqrt{1-\beta(u)^2}}$, where $\beta(u) = \frac{u}{c}$.
- 4-velocity $U^\mu \equiv \dfrac{dx^\mu}{d\tau} = \gamma(u)(c, \vec{u})$
- 4-acceleration $A^\mu \equiv \dfrac{dU^\mu}{d\tau} = \gamma(u)^2 \left[\gamma(u)^2 \dfrac{\vec{u}\cdot\vec{a}}{c}, \vec{a} + \gamma(u)^2 \dfrac{\vec{u}\cdot\vec{a}}{c^2}\vec{u}\right]$
- 4-momentum $P^\mu = mU^\mu$
- For a Lagrangian L, the action is $S \equiv \int_{t_i}^{t_f} L(q(t), \dot{q}(t), t)\, dt$
- The Euler-Lagrange EoMs are (for independent generalized coordinates)

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

PH3101, Spring 2024

End - Sem

DPS, IISER Kolkata

Instructor: Dr. Rajesh Kumble Nayak
10:00 AM, 02 November, 2024.
Duration 150 Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!

Q - 1(20 Marks)
For two functions $f(q,p,t)$ and $g(q,p,t)$, where (q,p) are canonical variable and tis time, show that,

$$\frac{d}{dt}[f,g] = \left[\frac{df}{dt}, g\right] + \left[f, \frac{dg}{dt}\right],$$

Here, $[f,g]$ is a Poisson bracket $w.r.t$ variables (q,p).

Q - 2(20 Marks)
For the Hamiltonian,

$$H = q_1p_1 - q_2p_2 - aq_1^2 + bq_2^2,$$

with a and b are constants,

1. Show that for a time-independent function of canonical variables, $G(p,q)$, if G is constant of motion, then
$$[G,H] = 0\,,$$
where is $H(p,q)$ is Hamiltonian.
2. Show that,
$$F_1 = \frac{p_1 - aq_1}{q_2}, \qquad \text{and} \qquad F_2 = q_1q_2\,,$$
are constants of motion.
3. Can you find more conserved quantities using Jacobies-identity?
Jacobies-identity look's like, $:= [X,\,[Y,\,Z]] + [Y,\,[Z,\,X]] + [Z,\,[X,\,Y]] = 0$, where, $[X,Y]$ is a Poisson bracket $w.r.t$ variables (q,p).

Q - 3(20 Marks)
Consider the Kepler problem, i.e. $V(r) = -\frac{k}{r}$.

1. Find the effective radial equation of motion.

P.T.O

2. For a bound orbit, the distance of closest approach (r_{min}) and farthest passage (r_{max}) of the planet is determined by the condition $\dot{r} = 0$. Find r_{min} and r_{max} in terms of E and L_z.
3. Mean distance is the length of semi-major axis, $a = (r_{min} + r_{max})/2$ and show that it depends only on the Energy of the particle!
4. For an artificial satellite, how conserved quantities such as E and L_z can be used to change the orbital parameters?

Q - 4(20 Marks)

For the Hamiltonian given below:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$$

1. Find an explicit canonical transformation $(q, p) \rightarrow (Q, P)$ such that Q is cyclic
2. Find the new Hamiltonian K and conserved momenta
3. Express the solution for EOM in canonical coordinates (q, p)

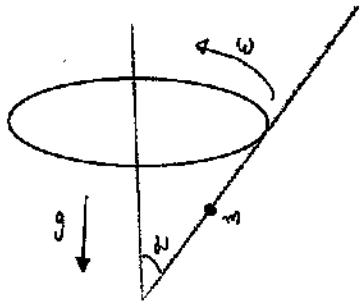
Q - 5(20 Marks)

The angular momentum is given by $\vec{L} = \vec{r} \times \vec{p}$ compute the component of angular momentum in Cartesian coordinate system $\vec{L} = (L_x, L_y, L_z)$, where $\vec{p} = m\vec{v}$, in terms of suitable canonical variables (q, p).

1. Compute the Poisson bracket $[L_x, L_y]$, $[L_y, L_z]$ and $[L_z, L_x]$ for the suitable canonical variables.

Q - 6(20 Marks)

A bead of mass m slides on a long straight wire, making an angle α with, and rotates with constant angular velocity ω about, the upward vertical. As shown in the figure below. Gravity g acts vertically downwards!



1. Choose an appropriate generalised coordinate and find the Lagrangian.
2. Write down Euler-Lagrange's equation of motion as the second-order ODE in generalised coordinates
3. Obtain the equation of motion from the energy conservation.
4. Are the equations of motion obtained from (1) and (2) are same? If not, why?

– The End –

6/12/24

PH3102 Quantum Mechanics **End-semester Examination**

Instructor: Dr. Siddhartha Lal **Autumn Semester, 2024**

Instructions: Answer all questions for full marks. Put your name and roll no. on both Question paper as well as the Answer script, and hand in both of them.

Date: 6th December, 2024 **Full marks: 50** **Time: 2.5 hours**

---

1 Squeezed states. [7 marks]

For the 1D simple harmonic oscillator, define an operator $Q = \mu a + \nu a^\dagger$, where a and $a^\dagger$ are the lowering and raising operators respectively:

$$a = \sqrt{\frac{m\omega}{2\hbar}}\left(x + \frac{ip}{m\omega}\right) \ , \ a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(x - \frac{ip}{m\omega}\right) \ , \ \left[a, a^\dagger\right] = 1 \ . \tag{1}$$

and where the complex numbers μ and ν satisfy the constraint $|\mu|^2 - |\nu|^2 = 1$.

(a) Show that $\left[Q, Q^\dagger\right] = 1$. [1 mark]

(b) Express the operators a and $a^\dagger$ in terms of Q and $Q^\dagger$. [1 mark]

(c) Now, consider an eigenstate $|\alpha\rangle$ of the operator Q, such that $Q|\alpha\rangle = \alpha|\alpha\rangle$ where $\alpha \in \mathcal{C}$. For the state $|\alpha\rangle$, and using the result of part (a), compute the expectation values $\langle x\rangle$ and $\langle x^2\rangle$ for the position operator. [3 marks]

(d) Show that $\Delta x = \sqrt{\frac{\hbar}{2m\omega}}|\mu - \nu|$, where $|z| = \sqrt{zz^*}$ for a complex number z. [2 marks]

Note that such a state is called a "squeezed state" if Δx is less than its "natural" value $\sqrt{\frac{\hbar}{2m\omega}}$.

2 Density matrix and the Liouville–von Neumann equation. [12 marks]

Consider an arbitrary state $|\psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle$ for a spin-1/2 object, where the states $|\uparrow\rangle$ and $|\downarrow\rangle$ are the eigenstates of σ^z with eigenvalues $1/2$ and $-1/2$ respectively. Such a state is placed in a magnetic field B, so that it evolves according to the Hamiltonian

$$H = B\sigma^z \ . \tag{2}$$

(a) Use the time evolution operator to find the time-evolved state $|\psi(t)\rangle$ at time t. [2 marks]

(b) Find the corresponding density matrix $\rho(t)$ at time t, where a density matrix is defined as $\rho = |\psi\rangle\langle\psi|$. [3 marks]

(c) Calculate the commutator $[H, \rho]$ between $\rho(t)$ and the Hamiltonian H. You can use the 2×2 matrix representation of ρ and H for this. [3 marks]

(d) By comparing the respective expressions, derive a relation between $d\rho/dt$ and $[H, \rho]$? This relation is called the Liouville–von Neumann equation. [Here, we consider the derivative of a matrix as simply element-wise differentiation.] [2 marks]

(e) What happens to the commutator when $b = 0$? What does this mean for the density matrix according to the Liouville–von Neumann equation? [2 marks]

3 Variational argument for Yukawa potential. [12 marks]

Consider the three dimensional attractive Yukawa potential:

$$V(r) = -A\frac{e^{-\lambda r}}{r} \ . \tag{3}$$

6/12/24

where the dimensional constant $A > 0$ and the inverse distance scale $\lambda > 0$ sets the range of the potential (beyond which the potential falls off very quickly).
(a) Compute the expectation value of the Hamiltonian $H = -(\hbar^2/2m)\nabla^2 + V(r)$ for the normalised variational wavefunction

$$\psi_\alpha(r) = \sqrt{\frac{\alpha^3}{\pi}} e^{-\alpha r} \ . \tag{4}$$

where the variational parameter α corresponds to a dimensionful constant α that sets a characteristic scale of the problem (e.g., for the case of $\lambda \to 0$, the Coulomb potential, $\alpha \propto a_0^{-1}$ the inverse Bohr radius and $A = e^2/4\pi\epsilon_0$). This will give you the variational energy $E(\lambda, \alpha)$. [6 marks]
Hint: Take the angular momentum quantum number $l = 0$ in the relation for the operator

$$\nabla^2 \equiv \frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \ . \tag{5}$$

(b) Minimise the variational energy $E(\alpha, \lambda)$ with respect to the variational parameter α to obtain the best variational estimate of the ground state energy E_{min}. State clearly the relation you obtain between α and λ as an outcome of the variational calculation. [4 marks]
(c) Show that there exists a relationship for the parameter λ such that E_{min} can accommodate bound states, $E_{min} < 0$. What does it indicate for the repulsive case of $A < 0$? [2 marks]

4 Anisotropic spin system. [12 marks]

(a) Two spins S_1 and S_2 are interacting with one other through the antiferromagnetic Ising spin exchange Hamiltonian

$$H_1 = J_\parallel S_1^z S_2^z \quad . \ J_\parallel > 0 \ . \tag{6}$$

Compute the eigenvectors and eigenvalues of the Hamiltonian H_1. [2 marks]
(b) Now consider two spins S_1 and S_2 that are interacting with one other through the antiferromagnetic anisotropic spin exchange Hamiltonian

$$H_2 = J_\perp \left(S_1^x S_2^x + S_1^y S_2^y\right) + J_\parallel S_1^z S_2^z \quad . \ J_\parallel > 0 \ . \ J_\perp > 0 \ . \tag{7}$$

Again, compute the eigenvectors and eigenvalues of the Hamiltonian H_2. How has the presence of the extra term in H_2, i.e., the term proportional $J_\perp$, affected the energy eigenvalues of H_2 with respect to those obtained for H_1? [7 marks]
(c) What happens to the eigenspectrum of H_2 for the special case of $J_\perp = J_\parallel$? [1 mark]
(d) Without conducting any explicit calculations, sketch pictorially the energy spectrum for the ferromagnetic exchange Hamiltonian with $J_\parallel, J_\perp < 0$ and (i) $J_\parallel \neq J_\perp$, and (ii) $J_\parallel = J_\perp$. [2 marks]

5 Perturbing the 2D Harmonic Oscillator. [7 marks]

Consider the two dimensional harmonic oscillator $H_0 = \frac{1}{2m}(p_x^2 + p_y^2) + \frac{m\omega^2}{2}(x^2 + y^2)$, to which we apply the perturbation $V = \lambda xy$.
(a) Write down the energy spectrum of H_0. For this, use the good quantum numbers that label the eigenstates of H_0. [2 marks]
(b) Compute the lowest order correction to the ground state of H_0 due to V. [2 marks]
(c) Compute the lowest order correction to the first excited state(s) of H_0 due to V. [3 marks]
Hint: Consider carefully any degeneracies in these states.

PH3103 Mathematical Methods of Physics
Autumn Semester - 2024
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

End-Semester Examination | Examination Date: 12/12/2024

1. (i) By explicitly calculating the integration (not using Residue theorem) over a convenient contour show that
$$\oint dz \frac{z^m}{z^n} = (2\pi i)\delta_{m+1,n} \ , \tag{1}$$
where m and n are integers. **Marks: 3**

   (ii) Interpret the result in terms of Cauchy's Residue Theorem. **Marks: 2**

2. Evaluate the following integral:
$$\oint \frac{1}{z-1}\frac{1}{(z-2)^2}dz \tag{2}$$
   (i) The closed contour is a circle $C_1 = 4e^{i\theta}$ for $0 \leq \theta \leq 2\pi$. **Marks: 4**

   (ii) Deform the contour into a contour $C_2 = Re^{i\theta}$ with $R \to \infty$, and estimate the value of the integration. Note that you are interested only in the limit of $R \to \infty$. **Marks: 2**

   (iii) Now, comment on two values in part (i) and (ii), in particular, why they are same/different! **Marks: 1**

3. Evaluate the following real integral using the complex integration techniques:
$$I = \int_0^\infty \frac{cos\ x}{q^2 + x^2}dx \tag{3}$$
   Marks: 6

4. Find out the Fourier transformation $g(x)$ of $f(k) = 1/(k^2 + m^2)^2$. **Marks: 9**

5. Rodrigues formula for the Legendre polynomials of n-th order is given by the following formula (discussed before the mid-semester)
$$P_n(x) = \frac{1}{2^n n!}\left(\frac{d}{dx}\right)^n (x^2-1)^n \ . \tag{4}$$
   (i) Show that it can be written as
$$P_n(z) = \frac{1}{2^n 2\pi i}\oint \frac{(z'^2-1)^n}{(z'-z)^{n+1}}dz' \tag{5}$$
   where the contour encircles the complex point z counterclockwise. **Marks: 3**

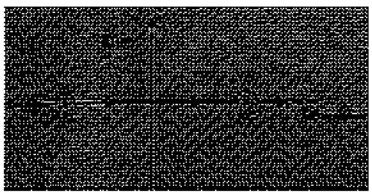
   (ii) From the above relation, show that $P_n(1) = 1$. **Marks: 3**

6. Show that eigenvectors of a unitary operator corresponding to different eigenvalues are orthogonal. **Marks: 5**

7. Obtain the retarded Green's function $G(X,T)$ (see the appropriate contour shown bellow) for the (1+1) dimensional wave operator

$$D = \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial t^2}\ , \tag{6}$$

where $X = x - x'$ and $T = t - t'$ (assuming translationally invariant boundary conditions). You need to find expressions in terms of integral over k, but do not need to evaluate it. But, surely, you need to perform the w integration! **Marks: 9**



8. The time-independent Schrodinger equation for a non-relativistic particle in a potential $V(r)$ can be written in the following form

$$(\nabla^2 + k^2)\psi(\vec{r}) = \lambda U(r)\psi(\vec{r}), \quad with \;\; U(r) = (2m/\hbar^2)V(r), \tag{7}$$

and λ is a 'small' parameter.

(i) Assuming free particle scattering state to be $\psi_0 = e^{i\vec{k}\cdot\vec{r}}$, write down the general solution of the above equation treating the R.H.S as the source term. For appropriate boundary conditions, it is given that

$$(\nabla^2 + k^2)\left(-\frac{e^{ik|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|}\right) = \delta^3(\vec{r}-\vec{r}'). \tag{8}$$

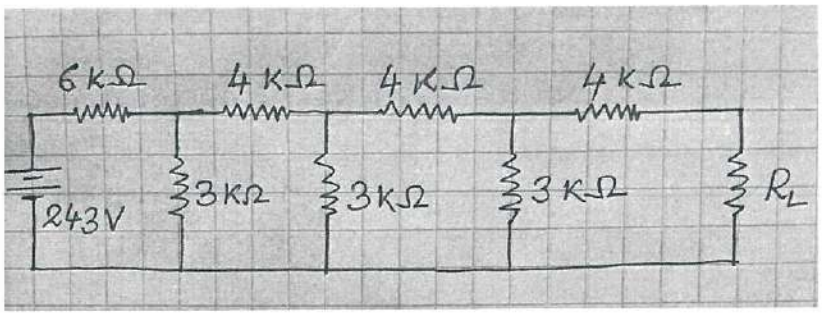
Marks: 2

(ii) Write down the approximate solution for $\psi(\vec{r})$ up to order λ. **Marks: 1**

End Semester Examination (Autumn 2024)
Course: PH3104 : Electrical Circuits and Electronics

Time: 2 hrs 30 mins | Total Marks: 50

1. (a) Calculate the load current in the following figure for the load resistance (R_L) of 3 kΩ: [3]



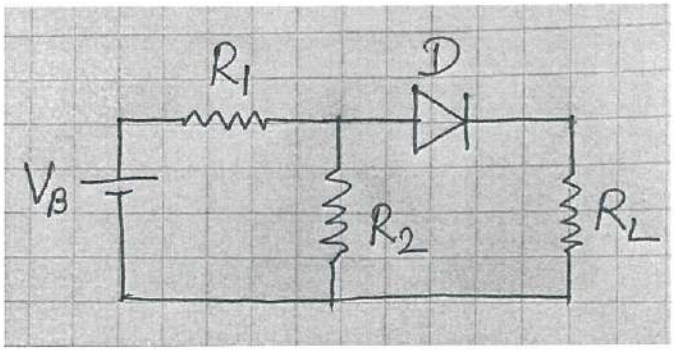
(b) Describe Half-voltage method for measuring R_{Th}. [2]

(c) Derive the expression of resonant frequency (f_r) and Bandwidth (BW) for a RLC parallel resonant circuit in terms of R, L and C. [2+3]

2. (a) The barrier potential developed across an open-circuited p-n junction aids the flow of minority carriers. Explain how the flow of charge is counter-balanced. [1]

(b) Draw the circuit diagram for a positive diode clipping circuit. Sketch the output waveform and discuss the observations. If the reference voltage is set to be zero and the Si diode is used in the circuit that is a non-ideal, then what will be the maximum positive and negative output voltages when an AC input voltage (Peak voltage of 50 V) is used? [1+(1+2)+1]

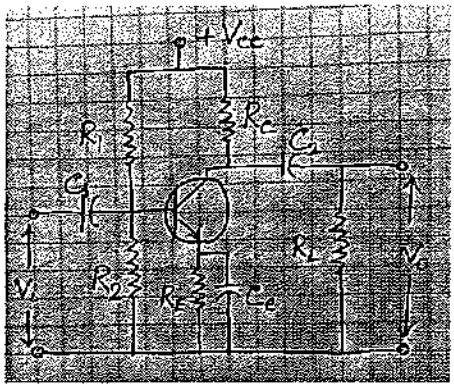
(c) Determine the current through the load resistor (R_L) in the following circuit, consider the Si diode used is a non-ideal diode. Also consider $V_B = 15$ V, $R_1 = 6$ kΩ, $R_2 = 4$ kΩ and $R_L = 2$ kΩ. [2]



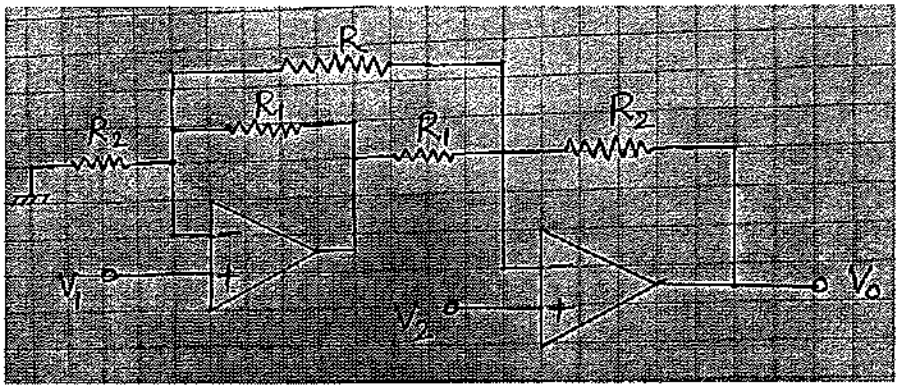
(d) Draw the circuit diagram for a AND logic gate and write down the truth table. [2]

3. (a) Define hybrid parameters for a transistor and draw the h-parameter equivalent circuit. Discuss how one can determine the h-parameters experimentally. [2+1+2]

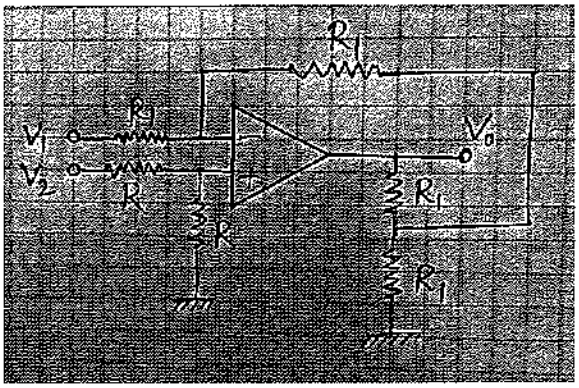
(b) In the CE amplifier circuit, $R_1 = 72$ kΩ, $R_2 = 8$ kΩ, $R_L = 2$ kΩ, $R_E = 700$ Ω and $R_C = 2$ kΩ and $V_{CC} = 15$ V. Assume $\beta = 120$ and $V_{BE} = 0.7$ V. Draw the dc load line and find the Q-point. [5]



4. (a) Discuss the differential amplifier with circuit diagram. [2]
   (b) The following circuit shows an instrumentation amplifier used as the input amplifier of many electronic instruments and to amplify the signals at the output of transducers for driving the indicator. Calculate the voltage gain ($A_V = \frac{V_o}{V_2 - V_1}$) of the amplifier. How can you vary the gain? [4+1]



   (c) Determine the output voltage V_o in the circuit below in terms of V_1 and V_2. [3]



5. (a) Give the logic symbol, Boolean expression and the truth table of a two-input NAND gate. [2]
   (b) Implement the Boolean function $Y = AB + \bar{A}C$ with NAND gates only. [2]
   (c) Simplify by Boolean algebra, $Y = \bar{A}B\bar{C} + \bar{A}C\bar{B} + A\bar{B}C + \bar{A}BC$ [2]
   (d) Simplify the Boolean expression $Y = \bar{A}BC + ABC + A\bar{B}C + AB\bar{C}$ using a Karnaugh map and draw the simplified logic circuit. [3 + 1]

Condensed Matter Physics(PII4101) End Sem Exam

time 2.5hr

1. Consider a 1d lattice of atoms with mass m. Within harmonic lattice approximation the potential between nearest neighbour atoms (at site i and $i+1$) is given by.

$$V = \frac{\kappa}{2}(x_{i+1} - x_i - a)^2 \tag{1}$$

a) For inter atomic potential $V = V_0[(\frac{\sigma}{r})^{12} - (\frac{\sigma}{r})^6]$, find out κ and lattice spacing a (3).
b)Write down the classical equations of motion of the displacement of atoms from the lattice points δx_i, using Eq(1). (3)
c)Obtain the dispersion of phonon modes by solving above equations. (5).

2. a) Consider 2d free electron gas and calculate number density as a function of chemical potential μ at a temperature T. (5).
b) Outline the steps to obtain the total energy of the gas at low temperature using Sommerfeld expansion. (You do not have to evaluate the final integral). (5)

3. a)Using Drude model write down the classical equation of motion of electrons in the presence of constant electric ($\vec{E} = E\hat{x}$), and magnetic field ($\vec{B} = B\hat{z}$). Average collision time is τ and electrons transfer complete momentum at each collision. (5).
b) Obtain the current density at steady state (3)
c) Find out Hall resistivity and write how it can be utilized to obtain the charge density. (2)

Next page

4. In 1d, the tight binding Hamiltonian is given by,

$$H = -t \sum_n [|n\rangle\langle n+1| + |n+1\rangle\langle n|] \tag{2}$$

where $|n\rangle$ represents localized state at n th lattice site.

a) Obtain the energy dispersion $E(k)$. (6)

b) Find out band width and effective mass. (3)

c)From $E(k)$, find out the density of states. (3)

d)For one electron per site (half filling), obtain the Fermi energy. (3)

5. a) In 2d lattice the unit vectors are $a_1 = a\hat{x}$ and $a_2 = a(\hat{x} + \hat{y})$. Find out the reciprocal lattice vectors.

b) Write down Braggs law for constructive interference from crystal. (2+2).

EndSem Exam - Autumn 2024
PH4102: Introductory Astrophysics

Answer Question 1 and any 4 out of the 5 remaining questions.

Note: Meaning of variables and symbols are same as we used in the class, unless stated otherwise. Clearly mention any assumption that you make while solving any problem.

Time : 150 min Marks: 50

(Given : $m_e = 9.1 \times 10^{-31}$ kg, $m_H = 1.67 \times 10^{-27}$ kg, $h = 6.62 \times 10^{-34}$ m^2 kg/s, $k_B = 1.38 \times 10^{-23}$ J/K, $G = 6.67 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, 1 AU = 1.5 $\times$ 10^8 km, $M_\odot$ = 1.99 $\times$ 10^{30} kg, $R_\odot = 7 \times 10^8$ m, $L_\odot = 3.85 \times 10^{26}$ W, c = 3 $\times$ 10^8 m/s, mass of Earth = 6 $\times$ 10^{24} kg)

1. a) The apparent magnitude of a variable star ranges from 7.1 to 7.8. Find out the ratio of maximum and minimum apparent brightnesses for this star. (2)

b) Write down the collisionless Boltzmann equation. Show that if $K(\vec{x}, \vec{v})$ is a constant of motion as a star moves around within a stellar system, a distribution function of the form f $(K(\vec{x}, \vec{v}))$ will give a time-independent solution of this equation. (2)

c) Inside Sun, around 0.71 % of hydrogen mass is converted to energy during hydrogen to helium fusion. Assuming this fusion reaction to be the dominant, calculate how much (in kg) hydrogen is fused per second to produce observed Solar luminosity of 3.9 $\times 10^{26}$ J/s? (2)

d) Draw the internal structure of a neutron star and mention the mass density values of each region. (2)

e) Describe last scattering surface. (2)

2. a) Inside dense core of a giant molecular cloud, criterion of gravitational collapse is satisfied and such a dense core is believed to be a site for star formation. Assuming free-fall, isothermal collapse, find out an expression for the free-fall timescale for such a cloud with constant density ρ_0. Clearly state any additional assumptions you make.

b) Assuming such dense cores are predominantly molecular hydrogen and number density of H_2 is $n_{H_2} = 10^{10}$ m^{-3}, find out numerical value of the free fall time scale. (8 + 2)

3. a) Derive induction equation of MHD:

$$\frac{\partial \vec{B}}{\partial t} = \vec{\nabla} \times (\vec{v} \times \vec{B}) + \eta \nabla^2 \vec{B},$$

with $\eta = \frac{1}{\mu_0 \sigma}$ is the resistivity.

b) Using this, show that under ideal condition, i.e., zero resistivity, one can prove magnetic flux conservation principle, namely,

$$\frac{d}{dt}\int_S \vec{B}.d\vec{S} = 0.$$

(4+6)

4. a) Describe how astronomers use virial theorem in stellar dynamics to get an estimate of the mass of a star cluster.

b) The Hubble Space telescope is in a nearly circular orbit, approximately 610 km above the surface of Earth. Estimate its orbital period.

c) Communication and weather satellites are often placed in geosynchronous orbits above Earth. These are orbits where satellites can remain fixed above a specific point on the Earth surface. At what altitude must these satellites be located?

(3+3+4)

5. a) Consider a ^{12}C nucleus of radius $r = 3 \times 10^{-15}$ m and compute its mean density.(assume mass of proton and neutron to be same.)

b) Pulsars are believed to be rapidly rotating neutron stars. Assuming the pulsar star to be a rigid sphere, show that a millisecond pulsar must have nuclear densities.

(3+7)

6. Consider a hypothetical star of radius R, with density ρ that is constant, i.e., independent of radius. The star is composed of a classical, non-relativistic, ideal gas of fully ionized hydrogen.

a) Solve the equations of stellar structure for the pressure profile, $P(r)$, with the boundary condition $P(R) = 0$.

b) Find the temperature profile, $T(r)$.

c) Assume that the nuclear energy production rate depends on temperature as $\epsilon \sim T^4$. At what radius does ϵ decrease to 0.1 of its central value, and what fraction of the star's volume is included within this radius? (3+2+5)

Non-Linear Dynamics PH4104

End-Semester Examination 2 December 2024

Duration: 2.5 hours. Full Marks: 50 Attempt all questions

Problem 1

Solve the PDE: $2\,x\,t\,u_x + u_t = u \qquad -\infty < x < +\infty, \quad t \geq 0$

Given that the initial condition is $u(x,0) = x$. **Marks: 10**

Problem 2

Define a suitable nonlinear transformation and obtain a general solution of: $u_t - a\,u_{xx} + b(u_x)^2 = 0.$

Given that the initial condition is $u(x,0) = f(x)$. **Marks: 5+7**

Problem 3

Consider a reaction-diffusion system:
$$\begin{aligned} u_t &= D u_{xx} + u^2 s - u \\ s_t &= s_{xx} + \mu\,(1 - u^2 s) \end{aligned}$$

(a) Obtain the condition for Hopf bifurcation

(b) Find the value(s) of the diffusion constant at which Turing bifurcation can occur. **Marks: 7+7**

Problem 4

(a) Calculate an exact value of the Lyapunov exponent for the logistic map $x_{n+1} = f(x_n) = 4x_n(1 - x_n)$.

[Use the topological conjugacy to tent map]

(b) Now couple two copies of the above logistic map as: $\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} 1-\epsilon & \epsilon \\ \epsilon & 1-\epsilon \end{pmatrix} \begin{pmatrix} f(x_n) \\ f(y_n) \end{pmatrix}.$

Find the critical coupling ϵ_c for the onset of synchronization. **Marks: 7+7**

Anandamohan
2/12/2024

PH4105 End Semester Examination

Total marks : 50 **Time : 150 mins**

*Answer **Question number one** and any **two** questions from the rest*

Q 1) Answer any five $(5 \times 4 = 20)$

a) Prove that the Lie bracket $[X, Y]$ of two vector fields $X, Y \in \mathfrak{X}(M)$, obey

$$[fX, gY] = fg[X, Y] + f(Xg)Y - g(Yf)X$$

for any smooth functions f and g on a manifold M.

b) If (U, ϕ) and (U', ϕ') are chart on M with coordinate functions $(x^1, \ldots, x^n)$ and $(x'^1, \ldots, x'^n)$respectively and if a 1-form α can be written as $\alpha_i dx^i$ and$\alpha'_i dx'^i$ in U and U' respectively, derive the relation between α_js and α'_i s in $U \cap U'$.

c) If ω and η are k- and l-forms, respectively, then

(i) prove that

$$\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$$

(ii) prove that

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$$

You can use the coordinate expansion for the forms.

d) Determine all the metric connection components (Christoffel symbols) corresponding to the metric tensor

$$g = \frac{1}{y^2}(dx \otimes dx + dy \otimes dy)$$

on the manifold $M \equiv \{(x, y) \in \mathbb{R}^2 : y > 0\}$.

e) Prove that the Riemann curvature tensor obeys

$$R(X, Y)(fZ) = fR(X, Y)Z$$

where $X, Y, Z \in \mathfrak{X}(M)$, and $f \in C^\infty(M)$.

f) From the Cartan structure equations prove that

$$d\mathcal{R}^i{}_j = \mathcal{R}^i{}_k \wedge \Gamma^k{}_j - \Gamma^i{}_k \wedge \mathcal{R}^k{}_j$$

g) For a rigid basis prove that

$$\Gamma_{ijk} + \Gamma_{jik} = 0$$

Q 2) The exterior derivative of a k-form, ω,is defined by

$$(k+1)\, d\omega\,(X_1, \ldots, X_{k+1}) \equiv \sum_{i=1}^{k+1} (-1)^{i+1} X_i \left(\omega\left(X_1, \ldots, \hat{X}_i, \ldots, X_{k+1}\right)\right)$$
$$+ \sum_{1 \le i < j \le k+1} (-1)^{i+j} \omega\left([X_i, X_j], X_1, \ldots, \hat{X}_i, \ldots, \hat{X}_j, \ldots, X_{k+1}\right)$$

where $X_1, \ldots, X_{k+1} \in \mathfrak{X}(M)$.

a) Prove that $d\omega$ is completely antisymmetric.

b) Show that it is $\mathbb{R}$-multilinear.

c) Show that it is C^∞-multilinear.

d) Prove from this definition that

$$d(f\omega) = f d\omega + df \wedge \omega$$

where $f \in C^\infty(M)$. $(6+2+3+4)$

Q 3) a) Prove the two Cartan Structure equations

$$\mathcal{T}^i = d\theta^i + \Gamma^i{}_j \wedge \theta^j$$
$$\mathcal{R}^i{}_j = d\Gamma^i{}_j + \Gamma^i{}_k \wedge \Gamma^k{}_j$$

b) Consider a manifold $M = \{(x, y) \in \mathbb{R}^2 : y > 0\}$ covered by the single chart (M, id). If the only nonzero connection coefficients are $\Gamma^1_{11} = \Gamma^1_{22} = \frac{1}{y}$, $\Gamma^2_{12} = -\frac{1}{y}$, calculate $T^i{}_{jk}$ and $R^i{}_{jkl}$ using the Cartan structure equations.

c) Prove that for a torsion free connection

$$\mathcal{R}^i{}_j \wedge \theta^j = 0$$

What does this imply for the curvature tensor components? $(6+5+(3+1))$

Q 4) Consider a 1+1 dimensional space-time with the metric given by

$$g = e^{2\mu(x)} dx \otimes dx - c^2 e^{2\lambda(x)} dt \otimes dt$$

where μ and λ are functions of x alone. It is obvious that $\theta^1 = e^{\mu(x)} dx$ and $\theta^0 = c e^{\lambda(x)} dt$ provide a rigid basis for 1-forms, *i.e.* a basis in which the components of the metric tensor are constant.

a) Find the nonzero Ricci rotation coefficients Γ_{ijk} for this basis.

b) Determine the nonzero $R^i{}_{jkl}$ in this basis.

c) Find the components of the Ricci tensor.

d) Determine the form of the functions $\mu(x)$ and $\lambda(x)$ for which the metric is Ricci flat, i.e. the Ricci tensor vanishes. $(4+4+2+5)$

Some Important Definitions/Relations

- **Connection coefficients :** $\nabla_{e_i}(e_j) = \Gamma^k{}_{ji} e_k$
- **Ricci rotation coefficients :** $\Gamma^i{}_{jk}$ or Γ_{ijk} in a rigid basis.
- **Torsion :** $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$
- **Curvature :** $R(X,Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}$
- **Connection 1-forms :** $\Gamma^i{}_j(X) \equiv \theta^i(\nabla_X e_j)$, $\Gamma^i_{jk} \equiv \Gamma^i{}_j(e_k)$
- **Torsion 2-forms :** $T^i(X,Y) \equiv \frac{1}{2}\theta^i(T(X,Y)) = \frac{1}{2} T^i{}_{jk} \theta^j \wedge \theta^k$
- **Curvature 2-forms :** $\mathcal{R}^i{}_j(X,Y) \equiv \frac{1}{2}\theta^i(R(X,Y)e_j) = \frac{1}{2} R^i{}_{jkl} \theta^k \wedge \theta^l$
- **Ricci tensor :** $R_{ij} \equiv R^k{}_{ikj}$

Date : 10th Dec, 2024 End-semester Exam

PH4106 : Quantum Field Theory

Duration: 14h00 to 16h30

Roll No: Name:

Time management: 1 marks problem should roughly take you 3.5 minutes.

Questions 1 – 3 are to be written in the answer book and question 4 – 8 on this question paper itself.

1. **Yukawa Theory:** Consider the a theory of a real scalar ϕ and a Dirac field ψ with a Yukawa interaction and scalar self interaction, given as 8 / (28 min)

$$\mathcal{L}_1 = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{m^2}{2}\phi^2 - \frac{\lambda}{6}\phi^4 + \bar{\psi}(i\gamma^\mu\partial_\mu - M)\psi - \frac{g}{2}\bar{\psi}\psi\phi$$

   (a) Write down the full set of Feynman rules for matrix elements in momentum space. 4

   (b) Consider the process $\phi\phi \to \bar{\psi}\psi$. Draw lowest order Feynman diagram(s) for the process and write down the corresponding matrix elements. 4
   (Only write the expressions for matrix-elements, no squaring or evaluation needed)

2. **Auxilary Field:** Consider a real fields ϕ and a real *auxilary* field Φ (i.e. field without kinetic term) described by the Lagrangian density 6 / (21 min)

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{m^2}{2}\phi^2 - \frac{M^2}{2}\Phi^2 - g\,M\,\Phi\phi^2$$

   (a) Obtain equations of motion (EoM) for two fields ϕ and Φ. 1

   (b) Obtain the stress-energy tensor $T^{\mu\nu}$ for the above Lagrangian density. 1

   (c) Use the algebraic EOM of Φ to elliminate Φ from the differential EoM of ϕ. Write down the effective Lagrangian density $\mathcal{L}_e(\phi, \partial_\mu\phi)$ that will give this new EoM of ϕ. 2

   (d) Obtain $T_e^{\mu\nu}$ for this new Lagrangian density $\mathcal{L}_e$. Compare $T_e^{\mu\nu}$ with the $T^{\mu\nu}$ with Φ substituted in terms of ϕ using the above EoM and comment on your findings. 2

3. **QED and Yukawa interaction:** Consider a gauge invariant theory of a fermion ψ with charge 1 unit, another fermion χ with charge 2 units and a complex scalar ϕ with charge 1 unit and a gauge boson. 14 / (50 min)

   (a) Write down the Lagrangian density for this gauge theory along with gauge transformation rules. 2

   (b) Expand the co-variant derivative and write all the terms explicitly. 3

   (c) Add a Yukawa like term, i.e. *anti-fermion – fermion – scalar* term to the above Lagrangian such that it respects the gauge invariance. 1

   (d) Write down the momentum space Feynman rules for Green's function for this theory including the Yukawa like term. 4

   (e) Which of the above rules needs modifications if we want to have Feynman rules for the matrix elements in the momentum space. Write down those modified rules for all the particles in the theory. 4

Time saver section: Write all the answers to the following questions on this page itself.

---

4. **Projection operator:** Mark the following properties of a general projection operator P as True or False. 2 / (4 min)

   (a) $P^{-1} = P$ True / False

   (b) $P^2 = \mathbb{I}$ True / False

   (c) $P^2 = P$ True / False

   (d) $P^T = P$ True / False

5. **Chirality and bi-linears:** Mark all the fermion bi-linear listed below that connect opposite chiralities of the fermions as True else False. 2 / (4 min)

   (a) $\bar{\psi}\gamma^5\psi$ True / False

   (b) $\psi\sigma_{\mu\nu}\psi$ True / False

   (c) $\bar{\psi}\gamma^\mu\psi$ True / False

   (d) $\bar{\psi}\gamma^5\gamma^\mu\psi$ True / False

6. **QED facts:** Consider the QED Lagrangian with massless fermion

$$\mathcal{L}_{QED} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}\ \not{D}\psi$$

   Mark all the following statments as True or False. 2 / (4 min)

   (a) Theory describes four degrees of freedom A^μ and four degrees of freedom ψ. True / False

   (b) Theory describes two degrees of freedom A^μ and four degrees of freedom ψ True / False

   (c) The theory has no interaction between left chiral and right chiral fermions. True / False

   (d) The number of photons are conserved in all of the processes in the theory. True / False

7. **Energy projection operator:** Given $\Lambda_\pm = (\pm\ \not{p} + m)/2m$ mark all the correction options as True else False. For the False statements, write down the correct expressions for the right hand side (RHS) of the equations. 3 / (8 min)

   (a) $\Lambda_+\ \Lambda_+ = \Lambda_+$ True / False

   (b) $\Lambda_+\ \Lambda_- = 0$ True / False

   (c) $\Lambda_-\ \Lambda_- = \mathbb{I}$ True / False

   (d) $\Lambda_\pm^{-1} = \Lambda_\mp$ True / False

8. **γ-matrices:** Mark all the correct options as True else False. For the False statements, write down the correct expressions for the RHS of the equations. 3 / (8 min)

   (a) $(\gamma^\mu)^\dagger = \gamma^\mu$ True / False

   (b) $(\gamma^5)^\dagger = \gamma^5$ True / False

   (c) $(\gamma^0)^\dagger = \gamma^0$ True / False

   (d) $(\gamma^5\gamma^\mu)^\dagger = \gamma^\mu\gamma^5$ True / False

IISER Kolkata
Department of Physical Sciences
Advanced Electricity, Magnetism and Optics, PH4107
Autumn Semester 2024
Final Examination, dated 06/12/2024
Max. Marks: 40

Q1. 4+3+3

(a) Find out the 1×4 Stokes vectors corresponding to the following 1×2 Jones vectors

(i) $\left[1 \quad \frac{(1+i)}{\sqrt{2}}\right]^T$, (ii) $[0.6 \quad 0.8\, e^{i\pi/3}]^T$

What kind of state of polarization does each of them represent?

(b) Using Jones algebra, show that two cascaded quarter-wave retarders with parallel fast axes are equivalent to a half-wave retarder. What would be the result if the fast axes are orthogonal?

(c) An anisotropic optical element transforms incident linearly polarized light (with Stokes vector $[1 \quad 1 \quad 0 \quad 0]^T$) into a right circularly polarized light (Stokes vector $[1 \quad 0 \quad 0 \quad 1]^T$). Identify the anisotropic optical element. What is the orientation angle of the anisotropy axis of the element with respect to the laboratory polarization axis? What would be the state of polarization, if the anisotropy axis of the element is rotated further by an angle 90°?

Q2. 5+5

(a) The scattering matrix (S) of a dielectric sphere of radius a ($a \ll \lambda$) is given by

$$S = K\begin{pmatrix} \frac{1}{2}(1+\cos^2\theta) & \frac{1}{2}(\cos^2\theta - 1) & 0 & 0 \\ \frac{1}{2}(\cos^2\theta - 1) & \frac{1}{2}(1+\cos^2\theta) & 0 & 0 \\ 0 & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & \cos\theta \end{pmatrix}$$

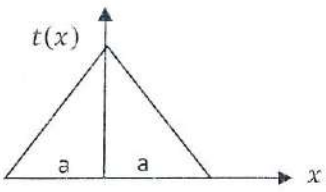
Determine the angular (θ) dependence of the degree of polarization (DOP) of the scattered light for (i) horizontally, (ii) vertically and (iii) unpolarized incident light. Plot the DOP vs $\theta(0^o - 180^o)$ variation in each case. Comment on the behavior of scattering and DOP at $\theta = 90\ deg$.

(b) In the above problem, what happens to the angular variation of DOP for incident right circularly polarized light? Comment on the state of polarization of the scattered light at different angular regions - $0^o < \theta < 90^o$, $\theta = 90^o$, $90^o < \theta < 180^o$

Q3. 5+5

(a) Consider Fraunhofer (far field) diffraction from the object shown below, transparency distribution of which is given by

$$t(x) = 1 - |x|,\ |x| \le a$$
$$= 0\ otherwise$$



6/12/24

Can you write the function $t(x)$ as a convolution of two functions?

(b) Derive expression for the intensity of far field diffracted waves from the object as a function of the spatial frequency $\left(\nu_{x'} = \frac{x_f}{\lambda r}\right)$ of the object. Here, x_f is the spatial co-ordinate at the Fourier (diffraction) plane and r is the distance from the object center. Plot the corresponding spatial frequency distributions at the Fourier plane.

Q4. 3+4+3

(a) In which one among the following two cases, can the light scattering be appropriately described using the quasi-static approximation (QSA)?

(i) Diameter of scatterer d = 20 nm, refractive indices of the scatterer and the surrounding medium, n_s = 1.37, n_m =1.33 respectively, wavelength λ = 600nm. (ii) d = 500 nm, n_s = 1.37, n_m =1.33, λ = 600nm.

In which case would the Born approximation be valid but not the QSA?

(b) Consider the following results of light scattering experiment. The measured scattered light intensity (I_s) from the two scattering samples is observed to vary with wavelength (λ) as - $I_s(\lambda) \approx \lambda^{-1.5}$, and $I_s(\lambda) \approx \lambda^{-2.5}$, respectively. The average dimensions of the scatterers are larger in which of the two samples? Briefly explain your answer.

(c) Unpolarized light with intensity I_0 is passing through cascade of three linear polarizers with transmission axes at 0°, 30°, and 60°, respectively. What will be the intensity after the 3rd polarizer?

OR

Q5. 5+5

(a) Consider a metal sphere (permittivity $\epsilon_m(\omega)$) with radius a = 20 nm embedded in air (ϵ = 1). The wavelength λ_p corresponding to the plasma frequency of the metal is given to be 250 nm.

(i) Determine the wavelengths corresponding to the dipolar (λ_d) and quadrupolar (λ_q) plasmon resonances of the metal sphere.

(ii) Compare the dipolar and quadrupolar plasmon resonance wavelengths of the sphere embedded in air with that of the same sphere embedded in water (ϵ = 1.77). In which case would the plasmon resonance wavelengths be higher? Briefly explain the physical basis of the differences in the resonance wavelengths.

[Use Drude's ideal free electron model for dielectric permittivity of metal, Use the condition of resonances for the dipolar and the quadrupolar polarizabilities.]

ENDSEM EXAM:

PH4110, SOFT CONDENSED MATTER PHYSICS, Autumn 2024

MARKS: 30, TIME: 2:30 HOURS, 07 December 2024, 02:00 PM

THIS EXAM WILL HAVE WEIGHT 30/100 FOR THE COURSE. ANSWER ALL SEVEN QUESTIONS (35 MARKS). I WILL TAKE YOUR TOTAL FOR 30/100 (MEANS YOU CAN EARN BONUS MARKS BY ANSWERING ALL SEVEN QUESTIONS). GOOD LUCK!

1. Consider a binary mixture (a two-component system). Assume the components have the same specific volume. Show that you can write the canonical free energy as $F = F(N_p, N_s, V, T) = Vf(\phi, T)$, where the symbols have usual meaning (total $N = N_p + N_s$; $\phi = N_p/(N_p + N_s)$ is the volume fraction for the p-component). Further, derive the common-tangent law for the coexistence of two phases in terms of ϕ.

[5 marks]

2. In the lattice model of a binary mixture (solvent and solute have equal specific volume), show that the mean-field internal energy can be written as $E = (1/2)Nz\Delta\epsilon\phi^2$, where N, z, ϕ are, respectively, number of lattice points, lattice co-ordination number, and solute volume fraction, and $\Delta\epsilon = \epsilon_{pp} + \epsilon_{ss} - 2\epsilon_{ps}$, where $\epsilon_{pp}, \epsilon_{ss}, \epsilon_{ps}$ are interactions, respectively between, solute-solute, solvent-solvent, and solute-solvent pairs.

Then write the total free energy of the system as a function of ϕ.

[5 marks]

3. Draw the pressure-density phase diagram of a van der Waal's fluid. Include the co-existence curve and a few isotherms including the critical isotherm. Describe the path the system follows in the case of two experiments: 1) The gas with an average density lower than the critical density confined in a fixed volume is cooled from a very high temperature, 2) the temperature of the liquid at a fixed pressure is increased (starting from a $T < T_c$).

[5 marks]

4. Show that, in the thermodynamic limit, and $S(\mathbf{q}) = \langle n \rangle [1 + \int \langle n \rangle g(\mathbf{x}) e^{-i\mathbf{q}\cdot\mathbf{x}} d^d x]$ for homogeneous and isotropic fluids. Further, show that for ideal gas $S_{nn}(\mathbf{q}) = \langle n \rangle$. Symbols have usual meaning.

7/12/24

Arindam Kundagrami

[5 marks]

5. Calculate the scattering intensity profile for the following density: $\langle n(\mathbf{x})\rangle = \rho_0 \cos(q_0 x)\delta(y)\delta(z)$. Is the density and scattering profile same as or different from the smectic-A phase? Discuss.

[5 marks]

6. Consider the Landau free energy:

$$f(\phi) = a\phi^2 + b\phi^4 + c\phi^6.$$

Which values of a, b, and c are possible for a physical system?

a) Draw the free energy and find the equilibrium ϕ for all sets of a, b, and c you can think of (consider positive and negative values of the coefficients).

b) When both b and c are positive, and a can be either positive or negative, how is the solution different from the case when $c = 0$?

c) How is a tricritical point defined? Derive the point in this system in terms of a and b.

[5 marks]

7. Identify the dispersion relation for photons and electrons from the following:

$$\epsilon = \hbar\omega, \epsilon = \frac{\hbar^2 k^2}{2m_e}.$$

a) For visible light, $\epsilon \sim 1$eV. What are the length scales that can be probed by scattering of visible light?

b) Typically in soft matters, what is the length-scale one tends to probe? What will be the temperature corresponding to a beam of neutrons of wavelength 1 angstrom?

[5 marks]

Formulae you may find useful (but may not require to use all of them):

1. The first law of thermodynamics:

$$dE = TdS - PdV + \mu dN$$

2.

$$C_v = \left(\frac{\partial E}{\partial T}\right)_{N,V}, C_p = \left(\frac{\partial (E + PV)}{\partial T}\right)_{N,P}$$

Arindam Kundagrami
7/12/24

PH5103 (Adv. CMP) End-semester exam, 03 December, 2024

1. Consider an independent electron of charge e moving in two-dimension (on x-y plane) is subjected to a *crossed* electric and magnetic fields. For definiteless, consider the constant magnetic field $\vec{B}$ applied in the $\hat{z}$-direction is generated from a vector potential given in Landau gauge: $\vec{A} = (0, Bx)$, defined on x-y plane, and the constant electric field $\vec{E}$ is in the $\hat{x}$-direction. Write down the resulting Hamiltonian of the electron. Solve the Shrödinger equation for the above Hamiltonian to obtain the eigenvalues and eigenfunctions. You can express your results using standard notations: $\omega_c = eB/m$, $l_B^2 = \hbar/(eB)$ in SI units. Here, the eigen energies will depend on relevant quantum numbers. Taking the derivative of the energy eigenvalue with respect to appropriate "momentum" calculate the drift velocity of the electron. (Hint: you need to operate $(\hbar)^{-1}(\partial/\partial k)$ on energy eigenvalues with appropriate k). Interpret your result from this quantum mechanical calculation in terms of the standard knowledge from classical mechanics.

[points: 12]

2. Consider the Hamiltonian: $\hat{\mathcal{H}} = \varepsilon\left(\hat{c}_1^\dagger\hat{c}_1 + \hat{c}_2^\dagger\hat{c}_2\right) + \Delta\left(\hat{c}_1^\dagger\hat{c}_2^\dagger + \text{h.c.}\right)$, where the parameters $\varepsilon > 0$ and $\Delta > 0$, and $\hat{c}_1^\dagger(\hat{c}_1)$ and $\hat{c}_2^\dagger(\hat{c}_2)$ are (spinless) fermionic creation (annihilation) operators satisfying canonical anticommutation relations. In order to diagonalize the Hamiltonia $\hat{\mathcal{H}}$, we introduce the canonical transformation:

$$\hat{d}_1 = u\hat{c}_1 - v\hat{c}_2^\dagger; \qquad \hat{d}_2 = u\hat{c}_2 + v\hat{c}_1^\dagger$$

where u and v are real numbers. (a) Use the requirement that the d-operators should also satisfy Fermionic canonical anticommutation relations and obtain the relation between u and v. (b) Assuming that $\hat{\mathcal{H}}$ is diagonalized upon expressring it in terms of d-operators, obtain the relationship involving u, v, ε and Δ which satisfies the diagonalized form of $\hat{\mathcal{H}}$. Finally express $\hat{\mathcal{H}}$ in the diagonal form:

$$\hat{\mathcal{H}}_{\text{diag}} = A\left(\hat{d}_1^\dagger\hat{d}_1 + \hat{d}_2^\dagger\hat{d}_2\right) + B$$

Here, B is an operator independent term. Obtain A and B in terms of ε and Δ. (c) Determine the energy and degeneracy of each energy level of $\hat{\mathcal{H}}$.

[points: 12]

3. Consider the inverse Bogoliubov transformation,

$$\gamma_{\vec{k},\sigma} = u_{\vec{k}}^* c_{\vec{k},\sigma} - \sigma v_{\vec{k}} c_{-\vec{k},-\sigma}^\dagger$$

Show that if the operators $c_{\vec{k}\sigma}$ and $c_{\vec{k}'\sigma'}^\dagger$ obey standard fermion anti-commutator relations, then the operators $\gamma_{\vec{k}\sigma}$ and $\gamma_{\vec{k}'\sigma'}^\dagger$ also obey these relations, provided, $|u_{\vec{k}}|^2 + |v_{\vec{k}}|^2 = 1$. Note the notations: When spin-subscript σ of operators (e.g., $c_{\vec{k}\sigma}$ or $\gamma_{\vec{k}\sigma}$) assumes spin index $\uparrow$ (or $\downarrow$), the corresponding in-line expression σ on the right hand side of the above transformation equation takes the value $+1$ (or -1) respectively. Also, the subscript $-\sigma$ of operators (e.g., $c_{-\vec{k},-\sigma}$) denotes a spin opposite to that of σ, in the above relation .

[points: 6]